
AWS VPC Endpoints – Complete Guide

What Are VPC Endpoints?

VPC Endpoints allow your AWS resources in a **private VPC** to securely connect to **AWS services without using the public internet**.

 “Private connectivity to AWS services from your VPC – no need for Internet Gateway, NAT, or public IPs.”

Types of VPC Endpoints

Endpoint Type	Description	Used For
 Interface Endpoint	ENI (Elastic Network Interface) in your subnet that connects to a supported AWS service	Services like S3, DynamoDB, SSM, Secrets Manager
 Gateway Endpoint	Adds a route in the route table for traffic to S3 or DynamoDB	Only for S3 and DynamoDB
 Gateway Load Balancer Endpoint (GLBEP)	Connects to a Gateway Load Balancer in another VPC	Advanced network inspection (firewalls, IDS, etc.)

1. Gateway Endpoints

- ♦ Used for **Amazon S3** and **DynamoDB** only.
- ♦ Adds a **prefix list route** to the **route table**, allowing private access.

Use Case

EC2 in a **private subnet** wants to upload to S3 **without NAT Gateway or Internet Gateway**.

Architecture

[EC2 in Private Subnet] ———▶ [S3] via Gateway Endpoint (No Internet)

2. Interface Endpoints

- ◆ Creates **ENIs (elastic network interfaces)** in your VPC.
- ◆ Supports **most AWS services** (SSM, Secrets Manager, CloudWatch, Lambda, etc.)
- ◆ Powered by **AWS PrivateLink**

Use Case

EC2 in private subnet uses **SSM**, or your app fetches secrets from **Secrets Manager**.

Architecture

[Private EC2] ———▶ [Secrets Manager] via Interface Endpoint (ENI)

- Appears as a **private DNS endpoint**
 - You don't need to use the public IP or internet
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How VPC Endpoints Work

For Gateway Endpoint (S3/DynamoDB)

1. Add **endpoint** to your VPC
2. Choose **S3/DynamoDB**
3. Add **route** to the route table:

- Destination: `p1-xxxxxxx` (prefix list for S3)
- Target: **Gateway Endpoint**

For Interface Endpoint

1. Creates an **ENI** with a **private IP**
2. Maps AWS service DNS to this private IP
3. Your VPC resources access service via private IP

Security

- VPC endpoints are **regional**
- You can attach **security groups** to interface endpoints
- You can use **VPC endpoint policies** to restrict access (IAM-style JSON policy)

```
{
  "Statement": [
    {
      "Principal": "*",
      "Action": "s3:GetObject",
      "Effect": "Allow",
      "Resource": "arn:aws:s3:::my-bucket/*"
    }
  ]
}
```

Hands-On Lab Idea (Gateway Endpoint for S3)

Goal:

EC2 in private subnet → Upload file to S3 using **VPC Gateway Endpoint**

Steps:

1. Create **VPC** with public + private subnets
 2. Launch **EC2 in private subnet** (no public IP, no NAT)
 3. Create **S3 bucket**
 4. Try uploading to S3 (will fail without internet)
 5. Go to **VPC** → **Endpoints** → **Create**
 - Service: `com.amazonaws.<region>.s3`
 - Type: **Gateway**
 - Attach to route table of **private subnet**
 6. Try uploading to S3 again 
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Hands-On Lab Idea (Interface Endpoint for Secrets Manager)

Goal:

EC2 in private subnet → Fetch secret from AWS Secrets Manager

Steps:

1. Launch EC2 in private subnet
2. Create **Secrets Manager** secret
3. Create **Interface Endpoint** for `com.amazonaws.<region>.secretsmanager`
 - Attach security group and subnet
4. SSH into EC2

5. Run Python/Boto3 script to access secret

Summary Table

Feature	Gateway Endpoint	Interface Endpoint
Services supported	Only S3 & DynamoDB	Almost all AWS services
Type	Route Table based	ENI-based
Traffic type	Routed	Private DNS / IP
Security	IAM policy	IAM policy + Security Group
DNS support	No	Yes
Use case	Private access to S3	Private access to SSM, Secrets Manager, etc.
